



Research paper

Enhancing glaucoma detection through multi-modal integration of retinal images and clinical biomarkers

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ABSTRACT

Glaucoma, a major cause of irreversible blindness globally, often progresses without early symptoms, making prompt and precise detection vital. This paper introduces a multi-modal glaucoma detection system that combines advanced deep learning architectures to analyze retinal images and clinical biomarkers. We developed three hybrid models: the first blends Vision Transformers (ViT) with Convolutional Neural Networks (CNN), specifically Residual Networks (ResNet), for comprehensive feature extraction; the second uses Object-Window-Location Vision Transformer (OWL-ViT) with Residual Networks for enhanced global contextual insights; and the third employs a Hierarchical Vision Transformer using Shifted Windows (Swin Transformer) with Residual Networks, which demonstrated the best performance. The strengths of these models, broad contextual capture by ViT, localized detail extraction by CNNs, and refined granularity by Swin Transformer, thereby improving both feature representation and computational efficiency, make them well-suited for clinical use. The best-optimized system, featuring the Swin Transformer hybrid model, achieved an F1-score of 0.993 for glaucoma and 0.995 for non-glaucoma, with an overall accuracy of 99.4% on a dataset of 2874 new cases, correctly classifying 2857 of them, thus confirming its efficacy in enhancing early-stage glaucoma detection and significantly advancing over existing methods.

1. Introduction

Glaucoma remains a leading cause of irreversible blindness globally, primarily due to its insidious progression and late diagnosis. Traditional diagnostic methods, including intraocular pressure (IOP) assessments, optic nerve imaging, and visual field testing, are hindered by the need for advanced equipment and specialized skills, restricting their accessibility, especially in under-resourced settings (Fu et al., 2018). While these methods are effective, their reliance on substantial infrastructure and interpretative expertise limits their widespread utility.

Recent advancements in artificial intelligence (AI), particularly through deep learning (DL) and machine learning (ML), have introduced innovative approaches to glaucoma detection, focusing primarily on the analysis of retinal images (Gheisari et al., 2021). Studies by Chen et al. (2023) and Sharma et al. (2020) have demonstrated the potential of convolutional neural networks (CNNs) in identifying pathological changes in retinal structures indicative of glaucoma. These advancements represent significant strides in automated glaucoma screening; however, they often overlook the integration of crucial

clinical biomarkers, which could enhance diagnostic precision.

Building on these technological advances, this study proposes a novel multi-modal approach that not only harnesses the capabilities of Vision Transformers (ViT), an architecture adapted from natural language processing for image analysis, but also integrates these with traditional machine learning analysis of clinical biomarkers. This fusion model leverages the strengths of both image-derived and clinical data, enhancing diagnostic accuracy and reliability beyond the capabilities of single-modality methods.

Guided by critical research questions, this study seeks to rigorously evaluate the efficacy of the proposed multi-modal diagnostic tool. Specifically, it addresses two pivotal questions: First, can the integration of Vision Transformers for fundus images with traditional machine learning analysis of clinical biomarkers improve the accuracy and reliability of glaucoma detection compared to single-modality image analysis methods? Second, how does the proposed multi-modal diagnostic tool compare against traditional and current AI-based methods in terms of accessibility, accuracy, and cost-effectiveness for early glaucoma detection? By exploring these questions, the study aims to

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substantiate the potential of a combined approach to revolutionize glaucoma diagnostics, enhancing both the precision of detection and the availability of critical diagnostic services.

By addressing these questions, our approach addresses critical gaps in existing methodologies by providing a more holistic diagnostic tool that combines detailed image analysis with patient-specific clinical indicators. This integrated system is evaluated against traditional and current AI-based methods (Cueto et al., 2021), showcasing its potential to revolutionize clinical practices by offering a more accessible, accurate, and cost-effective solution for early glaucoma detection. The implications of such advancements are profound, promising not only to democratize access to essential diagnostic services but also to facilitate timely interventions that could significantly mitigate the risk of irreversible vision loss in glaucoma patients.

2. Related work

Recent advancements in detecting glaucoma have substantially leveraged deep learning techniques, specifically through the analysis of retinal fundus images. Among these advancements, significant contributions include varied methodologies that utilize convolutional neural networks (CNNs) for segmentation and classification tasks. Notably, attention U-Net architectures have been pivotal for precise segmentation of optic disc (OD) and optic cup (OC), essential features for diagnosing glaucoma. For instance, the works of Shyamalee and Meedeniya (2022a) demonstrated that using attention U-Net models combined with CNN architectures such as Inception-v3, VGG19, and ResNet50, resulted in highly accurate segmentation of the OD.

In addition to segmentation, classification-only approaches using similar CNN configurations have shown promising outcomes. Another work by Shyamalee et al. Shyamalee and Meedeniya (2022b) applied these architectures to classify glaucoma from fundus images, with Inception-v3 on the ACRIMA dataset, underscoring the model's effectiveness in distinguishing glaucomatous conditions. Furthermore, integrated models that combine segmentation and classification like the work of Shyamalee and Meedeniya (2022c) have been developed, enhancing diagnostic accuracy and reliability. Such models have employed modified CNN architectures to perform both tasks sequentially, with one instance showing that an attention U-Net with ResNet50 for segmentation followed by Inception-v3 for classification. These studies collectively highlight the dynamic capabilities of CNNs in improving glaucoma detection through advanced image processing techniques.

Building on existing methods, Elmoufidi et al. (2023) introduced a CNN with multiple inputs that employ Bi-dimensional Empirical Mode Decomposition (BEMD) for feature extraction from fundus images, coupled with a Support Vector Machine (SVM) classifier to assess glaucoma. This approach has shown high precision (average of 97.31%) across multiple datasets and underscores the potential of integrating traditional image processing techniques with deep learning to enhance diagnostic accuracies.

Further explorations by Elmoufidi et al. (2022) utilize deep multiple instance learning for glaucoma prevention using color fundus photography. By decomposing regions of interest with BEMD and extracting features through CNNs, their model achieved fairly high sensitivity and specificity, showcasing the robustness of deep learning in medical diagnostics.

Additionally, the recent works of Kashyap et al. (2022) using DenseNet-201 for feature extraction and U-Net for image segmentation demonstrate a streamlined approach for glaucoma classification with a test accuracy of 96.09%, albeit without model ensembling or multimodal data considerations. Velpula and Sharma (2023) advanced this by employing a fusion of multiple pre-trained CNN models, leveraging their combined strengths to achieve higher accuracy through a voting-based classifier fusion.

In another notable study, Santos (2023) combines CNN-extracted features with machine learning classifiers like Random Forest, SVM,

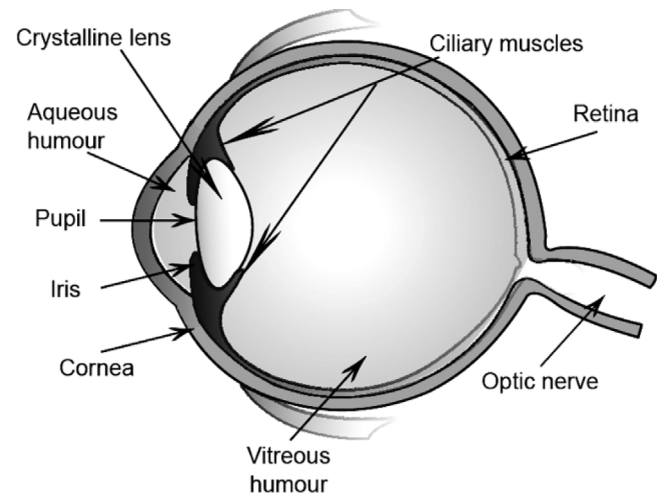


Fig. 1. Structure of the human eye (Practically, 2023).

and XGBoost. While effective at feature extraction, the subsequent classifiers may not fully capitalize on these features extracted by the CNN model, highlighting the need for models that better integrate the extracted data.

Diaz-Pinto et al. (2019) focused on employing several ImageNet-trained models for automatic glaucoma assessment, finding that Xception, with its depthwise separable convolutions, offered a balance between high accuracy and computational efficiency. However, the lack of multimodal data integration was noted as a drawback as this approach does not provide a holistic view of the patient's condition due to not considering their clinical biomarkers and other relevant health conditions.

Each of these methodologies, while advancing the field of glaucoma detection, presents limitations such as the need for extensive, representative datasets and the challenge of translating models effectively across different populations. Our study addresses these gaps by introducing three novel hybrid models that enhance diagnostic capabilities. These models integrate advanced deep learning architectures—specifically the Vision Transformer (ViT), Object-Window-Location Vision Transformer (OWL-ViT), and the Hierarchical Vision Transformer using Shifted Windows (Swin Transformer)—each paired with ResNet50, to analyze retinal images. Additionally, these models are equipped with sophisticated algorithms to interpret structured clinical data, providing a robust, multimodal approach that improves the accuracy and reliability of glaucoma detection. This comprehensive approach aims to enhance glaucoma detection's diagnostic accuracy and reliability, facilitating earlier interventions and potentially improving patient outcomes in the fight against this blinding disease. By synthesizing image-based and clinical insights, this model could significantly advance the precision of glaucoma screenings.

3. Overview of eye structure

The human eye is a complex organ essential for vision, featuring an intricate assembly of tissues and blood vessels that present challenges in theoretical study and clinical diagnosis. Fig. 1 illustrates the eye's detailed structure, emphasizing its complexity.

Our research focuses on the eye's fundus, which is located opposite the lens and comprises crucial anatomical components like the retina, optic disc, and macula (Fig. 2). These parts of the eye are critical for diagnosing diseases such as glaucoma, which our study primarily addresses using the three hybrid model architectures consisting of ViT (Alexey, 2020), OWL-ViT (Minderer et al., 2022), and Swin Transformer (Liu et al., 2021), each combined with ResNet50. Glaucoma can



Fig. 2. Example Fundus Image ([dataset] DeathTrooper, 2021).

lead to optic nerve damage and vision loss, intensified by increased intraocular pressure. We employ advanced imaging techniques like CLAHE and deep Vision Transformer models to enhance glaucoma detection and improve treatment outcomes.

4. Overview of deep learning architectures used

4.1. Neural networks architecture

Neural networks, a cornerstone of deep learning technology (Rosenblatt, 2021), are critically important when in-depth analysis is required for complex use cases due to their remarkable accuracy. A neural network consists of multiple neurons arranged in layers, where the architecture, including the number of layers and the connections between these layers—varies based on the specific task being evaluated and the neural network type utilized. Neurons in these networks are assigned weights and are interconnected through biases, which may be adjusted to suit the problem's dynamics. The fundamental architecture of a neural network is illustrated in Fig. 3.

The architecture depicted is a feed-forward network with fully connected layers, encompassing an input layer, several hidden layers, and an output layer. While numerous neural network types exist, such as Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs), our research employs an Encoder–Decoder architecture. This sophisticated convolutional network structure is particularly suited for biomedical image processing, allowing for detailed and complex analyses crucial for complicated conditions like glaucoma.

4.2. Convolutional neural networks

CNNs are renowned for their effectiveness in two-dimensional image processing with minimal preprocessing required. They are composed of two primary segments: feature extraction and classification. The feature extraction portion includes convolutional layers paired with activation functions, typically ReLU, and is complemented by pooling layers which help reduce spatial dimensions and computational complexity. The classification segment utilizes traditional fully connected layers. Unlike conventional machine learning techniques that require manual feature specification, CNNs autonomously learn and generate feature maps, making them exceptionally efficient for tasks involving complex image data, such as the analysis of retinal images for glaucoma detection.

4.3. Vision transformers (ViT)

Vision Transformers (ViT) represent a significant advancement in computer vision by adapting the transformer architecture traditionally used for language processing to image analysis tasks. Unlike Convolutional Neural Networks (CNNs), which process images through localized filters, Vision Transformers treat an image as a sequence of fixed-size patches, extracting features globally across patches through self-attention mechanisms. This approach allows ViTs to consider the entire context of an image, making them particularly effective for complex tasks such as detecting subtle indicators of glaucoma in retinal images. Each patch is embedded, combined with positional information, and processed through multiple layers of transformers that learn to focus on relevant parts of the image dynamically. This capability enables Vision Transformers to perform highly detailed and accurate image classification, directly benefiting tasks like medical image analysis where precision is crucial.

5. Our methodology

Fig. 4 illustrates our comprehensive methodology adopted for multi-modal glaucoma detection, encompassing both advanced image processing techniques and clinical data analysis. This visual representation provides a concise overview of the various stages involved in the study, starting from the initial image preprocessing, which includes green channel extraction and adaptive histogram equalization, to the sophisticated machine learning approaches utilizing Vision Transformers (ViT Alexey, 2020, OWL-ViT Minderer et al., 2022 and Swin Transformer Liu et al., 2021) combined with ResNet50 for image feature extraction and Random Forest classifiers for analyzing clinical biomarkers. The integrated features from these distinct modalities are then fused at a feature level, enabling the simultaneous exploitation of textural, structural, and clinical information to enhance diagnostic accuracy. This diagram serves as a guide through the complex workflow of our dual-modality approach, highlighting our novel fusion technique that aims to significantly improve early glaucoma detection rates with a test accuracy rate of 99.4%.

5.1. Dataset used

Our study used the Standardized Multi-Channel Dataset for Glaucoma (SMDG-19) ([dataset] DeathTrooper, 2021) as the foundational dataset for training our image-based hybrid model. The SMDG-19 ([dataset] DeathTrooper, 2021), comprising 19 consolidated public datasets (12,449 images), represents the largest publicly available repository of full-fundus glaucoma images. This comprehensive dataset not only includes the fundus images but also comes enriched with crucial image metadata such as optic disc and cup segmentation, blood vessel segmentation, and demographic details like age and sex associated with each instance. The dataset is designed to foster exploratory and open-ended research with versatile use cases, lacking predefined training, validation, or test splits to encourage novel experimental setups.

To accommodate effective machine learning workflows, the images in SMDG-19 ([dataset] DeathTrooper, 2021) have been standardized through a novel algorithm, which is meticulously designed to maximize the retention of foreground information. This standardization process involves cropping extraneous background, centering the fundus within the frame, padding any missing information from the original dataset, and resizing images to a uniform resolution of 512x512 pixels. This rigorous preparation ensures the dataset is optimally configured for advanced image processing tasks necessary for glaucoma detection. The glaucoma labels across the 19 datasets incorporated into the SMDG-19 ([dataset] DeathTrooper, 2021) are detailed in Table 1, highlighting the dataset's extensive scope and utility in our research.

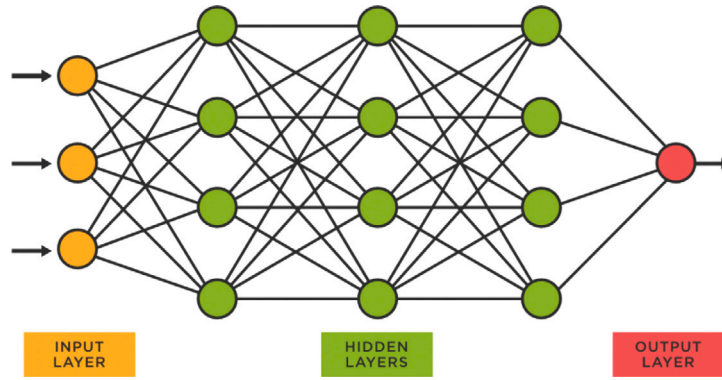


Fig. 3. Basic Neural Networks Architecture (MarkTechPost, 2022).

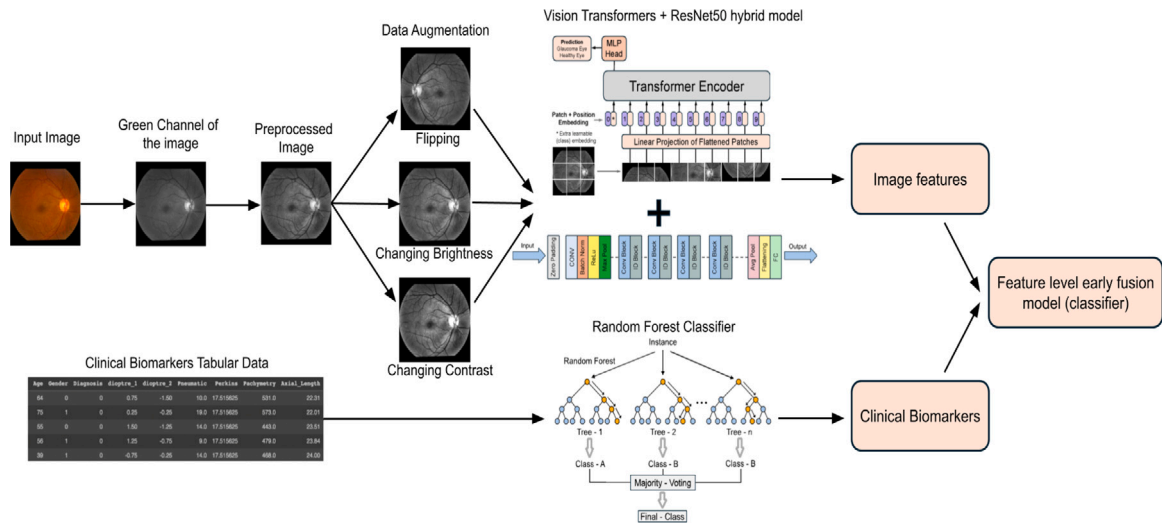


Fig. 4. Our Comprehensive methodology for multi-modal glaucoma detection.

Table 1
SMDG-19, A Standardized Fundus Glaucoma Dataset ([dataset] DeathTrooper, 2021).

Dataset	Non-Glaucoma (label: 0)	Glaucoma (Label: 1)	Suspect (label: -1)
BEH, Bangladesh Eye Hospital (Islam et al., 2021)	463	171	0
CRFO-v4 (Hassan et al., 2019, 2020)	31	48	0
DR-HAGIS (Holm et al., 2017)	0	10	0
DRISHTI-GS1-TRAIN (Sivaswamy et al., 2015)	18	32	0
DRISHTI-GS1-TEST (Sivaswamy et al., 2015)	13	38	0
EyePACS-AIROGS (De Vente et al., 2023)	0	3269	0
FIVES (Jin et al., 2022)	200	200	0
G1020 (Jain, 2022)	724	296	0
HRF ([dataset] Computer Science Department, Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), 2011)	15	15	0
JSIEC-1000 (Chundan, 2019)	38	0	13
LES-AV (Orlando et al., 2018)	11	11	0
OIA-ODIR-TRAIN ([dataset] Andrew MVD, 2020)	2932	197	18
OIA-ODIR-TEST-ONLINE ([dataset] Andrew MVD, 2020)	802	58	25
OIA-ODIR-TEST-OFFLINE ([dataset] Andrew MVD, 2020)	417	36	9
ORIGA-light (Zhang, 2022)	482	168	0
PAPILA (Kovalyk et al., 2022)	333	87	68
REFUGE1-TRAIN (Orlando et al., 2020)	360	40	0
REFUGE1-VALIDATION (Orlando et al., 2020)	360	40	0
sjchoi86-HRF (Chen, 2016)	300	101	0
Total	7499	4817	133

To analyze glaucoma patient biomarkers in our study, we utilized clinical data from the PAPILA dataset (Kovalyk et al., 2022), encompassing comprehensive clinical records for 244 patients. This dataset is organized into two distinct files corresponding to the right eye

(OD, Oculus Dexter) and the left eye (OS, Oculus Sinister), reflecting standard ophthalmological nomenclature. Each file is structured in a spreadsheet format, with rows representing individual patients and columns detailing a range of diagnostic and physiological data. The



Fig. 5. The red image represents the original fundus image from the SMDG-19 dataset, while the subsequent images illustrate the results after applying our preprocessing techniques.

fields in these tables include a unique patient identifier, patient age, gender (coded as 0 for male and 1 for female), diagnostic outcome (where 0 indicates a healthy eye, 1 denotes diagnosed glaucoma, and 2 signifies a suspicious case), refractive error, phakic/pseudophakic status (with 1 indicating removal of the crystalline lens and 0 its retention), intraocular pressure, pachymetry measurements, and axial length of the eye. Additionally, the mean defect in visual field testing is recorded, providing a crucial metric for assessing the functional impairment associated with glaucoma. This detailed clinical dataset allows for a multifaceted analysis of potential glaucoma indicators, thereby enhancing the predictive capabilities of our hybrid model.

5.2. Data preprocessing and augmentation

For proper detection and convenience, we ignore the Suspecting Glaucoma rows. We only focus on patients Diagnosed with Glaucoma and patients without Glaucoma.

5.2.1. Image data preprocessing

In our study, we implemented a specific technique for generating high-resolution retinal images, which is critical for the accuracy of glaucoma detection. While many groups spend extensive time creating high-resolution images, our technique involves a unique preprocessing pipeline that ensures maximum detail and clarity. This method enhances the visibility of critical retinal features, such as the optic disc and blood vessels, which are essential for accurate diagnosis. By applying this technique, we ensure that our models receive the best quality input data, significantly improving their performance in detecting glaucoma.

Our study utilizes a systematic approach to preprocess the retinal images sourced from the SMDG-19 dataset ([dataset] DeathTrooper, 2021), ensuring they are primed for the deep learning model. The preprocessing workflow begins with converting images from BGR to RGB color space to align with the standard input format for neural networks. Each image is then processed to isolate the green channel, which highlights the vasculature of the retina, an essential feature for detecting signs of glaucoma.

Following the extraction of the green channel, we apply Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance the contrast of the images. This method is particularly effective in enhancing the visibility of essential features in retinal images, such as the optic disc and blood vessels, which are crucial for accurate diagnosis. The transformation pipeline is applied to each image in our dataset, consisting of training, validation, and test sets. These are then resized to a uniform dimension of 512×512 pixels to facilitate consistent input to the neural network model. Fig. 5 compares the original image and the image after Data Preprocessing.

We also explored additional preprocessing techniques, such as histogram segmentation (Vijayalakshmi and Nath, 2023b), total variational decomposition (Vijayalakshmi and Nath, 2023a), and multilevel histogram equalization (Vijayalakshmi and Nath, 2022). We observed that these methods, although innovative, did not yield the same level of diagnostic improvement as our established method for the dataset that we had. For instance, the approach using total variational decomposition (Vijayalakshmi and Nath, 2023a) resulted in a few blurred images, which proved unsuitable for training our models with the dataset we had. Similarly, multilevel histogram equalization (Vijayalakshmi and Nath, 2022) provided a really good performance. However, it did not consistently outperform our original method in terms of clarity and feature enhancement for all images as it introduced some noises in some images.

In comparison, our original preprocessing pipeline not only maintained image integrity but also effectively enhanced crucial diagnostic features for our dataset. This was evidenced by superior model performance metrics, with our models achieving a test accuracy of 99.4% compared to the 90.327% achieved while using multilevel histogram equalization. These results underscore the effectiveness of our preprocessing method, making it a more reliable choice for the dataset we had for enhancing early-stage glaucoma detection.

To organize our data effectively, we created directories for each category within the training, validation, and test sets. Each directory contains images that have undergone the aforementioned preprocessing steps. We developed a script to automate the processing and categorization of images based on their glaucoma status, represented by labels: 0 for non-glaucoma, 1 for glaucoma. This structured approach not only streamlines the workflow but also ensures consistency and reproducibility in our experiments.

5.2.2. Image data augmentation

To mitigate the risk of overfitting and enhance the robustness of our model against variations in unseen data, we employ several data augmentation techniques during our training. Specifically, during our training, images undergo random horizontal flips and variations in brightness and contrast. The brightness is adjusted by up to 10%, and contrast is altered between 20% and 30%. These augmentations simulate different lighting conditions and imaging scenarios, thereby providing a more comprehensive training regime. Our approach ensures that our model can generalize better to diverse and unforeseen data inputs.

5.2.3. Clinical data preprocessing

To enhance the model's ability to predict glaucoma from clinical biomarkers, we utilized clinical data from the PAPILA dataset (Kovalyk et al., 2022), which consists of separate files for the right eye (OD,

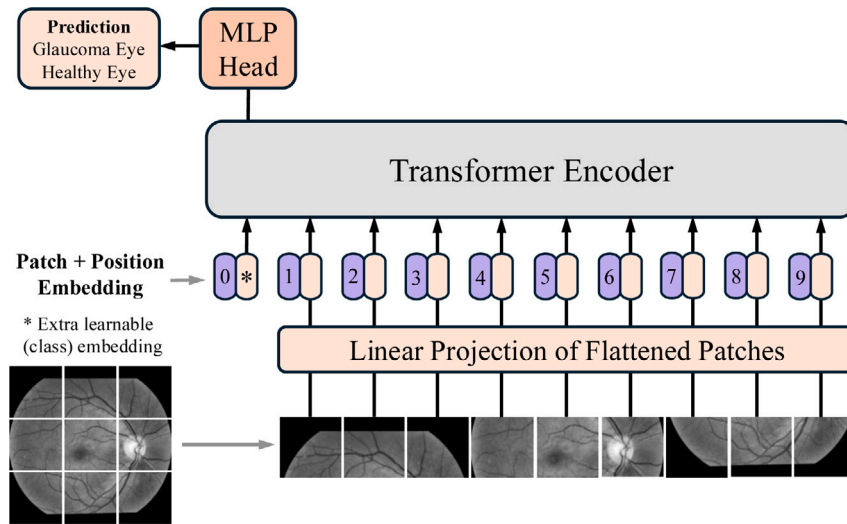


Fig. 6. Our Vision Transformer Methodology.

Oculus Dexter) and the left eye (OS, Oculus Sinister), containing comprehensive clinical data for 244 patients. Initially, we loaded this data into pandas dataframes, standardized the column headers using the first row of each dataset, and assigned a unique identifier for each patient. We then combined both dataframes into a single frame to unify the data structure, which facilitated the simultaneous processing of biomarkers from both eyes. In the pursuit of focusing on relevant biomarkers for glaucoma detection, we removed non-essential columns such as 'ID', 'astigmatism', 'Phakic/Pseudophakic', and 'VF_MD'. We also addressed missing values in crucial columns like 'dioptr_1', 'dioptr_2', 'Pneumatic', 'Perkins', 'Pachymetry', and 'Axial Length' by filling them with the mean of each column to maintain data integrity.

Furthermore, we filtered out entries under 'suspicious' diagnosis to ensure the training was focused on clear cases. To normalize the data distribution and improve the model's performance, we applied a StandardScaler transformation, which standardized the features by removing the mean and scaling them to unit variance. This scaling is crucial for models that assume normally distributed data. Finally, we split the processed and balanced data into training and test sets, allocating 80 percent of the data for training and 20 percent for testing. This split was strategically chosen to provide a comprehensive learning set while ensuring robust model validation on unseen data, essential for assessing the model's generalizability and effectiveness in glaucoma detection.

5.2.4. Clinical data augmentation

Given the imbalanced nature of medical datasets, particularly in the context of disease diagnosis where disease prevalence may be low, we employed the Synthetic Minority Over-sampling Technique (SMOTE) (Nitesh, 2002). This technique generated synthetic samples for the minority class in our dataset, thereby balancing the distribution between healthy and glaucoma-diagnosed patients. This balancing is crucial for preventing model bias towards the majority class.

5.3. Vision transformers

Vision Transformers (ViT) (Alexey, 2020) employ a novel approach that adapts the transformer architecture as seen in Fig. 6, initially designed for natural language processing, to the field of image analysis. This methodology treats images as sequences of patches, similar to how text is treated as sequences of words. This shift in perspective allows Vision Transformers to capture complex patterns and contextual relationships across an image, making them particularly suitable for the detailed analysis required in medical imaging, such as in glaucoma

detection.

Our approach with Vision Transformers starts by partitioning a retinal image into numerous fixed-size patches. By utilizing this technology, we format our image data into a sequence of patches. These patches are then flattened and linearly transformed into embeddings. This method prepares the image data in a format akin to textual data, allowing the transformers to apply their robust self-attention mechanisms effectively.

We overcome the non-inherent ability of transformers to recognize the order of input by adding positional encodings to the embeddings of the patches. Our results have shown that this step is crucial to preserve the spatial hierarchy of the retinal image features, such as the optic disc, blood vessels, and other structures relevant to identifying signs of glaucoma.

The transformer encoder consists of multiple layers, each with multi-head self-attention mechanisms and position-wise feed-forward networks. For glaucoma detection, these layers enable the model to focus on different areas of the retinal image, assessing intricate relationships and patterns such as changes in the optic nerve head or retinal nerve fiber layer, which are vital markers for diagnosing glaucoma.

After processing through the transformer encoder, the image representation is pooled, typically using a classification token learned during training. This pooled output captures the contextual information from the entire retinal image, facilitating the classification of the image as either indicative of glaucoma or not.

In glaucoma detection, the ability to analyze high-resolution fundus images comprehensively is imperative. We selected Vision Transformers (ViTs) to leverage their advanced capability to interpret extensive contextual information and detect complex patterns across the retina. This approach ensures that we have a comprehensive technique to obtain the best and most generalized images. By utilizing ViTs, we were able to discern subtle and diffuse signs of glaucoma more effectively, significantly improving the accuracy of our early glaucoma detection. This enhanced image analysis capability has facilitated earlier interventions and potentially better patient management, making our approach both robust and reliable.

5.4. Our hybrid architecture: Vision transformers plus ResNet50

In our pursuit of advancing glaucoma detection from retinal images, we implemented three hybrid model architecture that combines the strengths of Vision Transformers (ViT) and ResNet50 renowned for their performance in image analysis tasks. This combination enables our model to capture both macro and micro-level details within

the images, facilitating a comprehensive analysis of structural and morphological changes in the retina.

1. ViT and ResNet50 Hybrid: The first model employs the Vision Transformer architecture, which processes images as sequences of patches and applies self-attention mechanisms across the entire image. This method is particularly valuable for identifying intricate patterns essential for medical diagnostics.

2. OWL-ViT and ResNet50 Hybrid: The second architecture integrates the Object-Window-Location Vision Transformer (OWL-ViT) with ResNet50. OWL-ViT enhances the ability to process contextual relationships within the images, making it exceptionally suitable for highlighting relevant medical features such as the optic disc and blood vessels.

3. Swin Transformer and ResNet50 Hybrid: The third model utilizes the Swin Transformer, which uses shifted windows to manage the self-attention mechanism, allowing for scalable and efficient processing of large images. This approach is known for its capability to maintain high-resolution details through various layers of the network, which is crucial for detecting subtle glaucomatous signs in fundus photographs.

The Transformer segment leverages the Vision Transformer architecture's capability to process images as sequences of patches, applying self-attention mechanisms that consider relationships across the entire image. This feature is particularly crucial for highlighting intricate patterns essential for medical diagnostics. In our implementation, the Transformer is configured without the top classification layer (`include_top=False`), allowing us to customize the output to meet our specific diagnostic requirements. It is pre-trained on large datasets, equipping it with a robust initial understanding of visual features.

Complementing the Transformer, the ResNet50 architecture helps us extract localized features through its deep residual learning framework. This CNN model excels in identifying spatial hierarchies, which is vital for recognizing the subtle signs of glaucoma in fundus photographs. We configured ResNet50 similarly with (`include_top=False`) and setting its layers to non-trainable to preserve the integrity of the features learned from its pre-training on ImageNet.

Each model independently processes the input images, extracting distinct sets of features. The Transformer focuses on global contextual relationships, while ResNet50 targets specific textural and structural details. These outputs are then flattened and concatenated to form a comprehensive feature set combining both global and local perspectives.

After combining features, the model includes several fully connected layers with GELU activation functions to facilitate non-linear processing. A dropout layer is introduced to prevent overfitting by randomly omitting a subset of features during training. The final output layer, featuring a softmax activation function, classifies the inputs into glaucomatous and non-glaucomatous eye categories.

The model is compiled with an AdamW optimizer, adjusting learning rates based on weight decay, and uses a sparse categorical cross-entropy loss function tailored for our categorical targets. We conducted training for 30 epochs for our ViT hybrid model and over 10 epochs for both OWL-ViT hybrid model and Swin Transformer hybrid model as the results already converges due to the high complexity of the models. This process uses batches of images from our preprocessed and augmented dataset, enhancing the model's robustness against various image transformations.

Our hybrid approach has improved the model's accuracy in detecting glaucoma and significantly enhanced its ability to generalize across different representations of pathological features in retinal images. This has led to more reliable and earlier disease detection, with our initial tests showing a marked improvement in sensitivity and specificity compared to existing methods. Following implementation, we observed that our model could successfully detect early-stage glaucoma in previously undiagnosed or misdiagnosed patients using traditional examination techniques. This outcome has potential implications for improving patient management and treatment strategies, offering a promising avenue for reducing the progression and impact of glaucoma.

Table 2

ML models accuracy comparison.

Model	Accuracy	F1-Score
Logistic Regression	70.15%	0.70
Support Vector Machine	82.84%	0.83
Random Forest	94.03%	0.94
Gradient Boosting	92.54%	0.93
K-Nearest Neighbors	82.09%	0.82
Decision Tree	92.54%	0.93
Naive Bayes	75.37%	0.75

5.5. Models used for clinical data

To evaluate the effectiveness of clinical biomarkers in predicting glaucoma, we employed a suite of diverse machine learning classifiers, each known for its unique strengths in pattern recognition and predictive analytics. The classifiers tested include Logistic Regression (Cox, 1958), Support Vector Machine (SVM) (Cortes, 1995), Random Forest Classifier (Alghamdi and Alharby, 2019), Gradient Boosting (Friedman, 2001), K-Nearest Neighbors (KNN) (Fix, 1985), Decision Tree (Quinlan, 1986), and Naive Bayes algorithm (Kononenko, 1990).

Each model was trained using the scaled training dataset, which comprised a balanced representation of the classes by applying SMOTE (Nitesh, 2002) to the original skewed dataset. The training process focuses on fitting the models to discern patterns that could accurately predict the glaucoma status from clinical biomarkers.

We evaluated the performance of each classifier based on its accuracy in predicting the glaucoma status on a scaled test dataset. Additionally, we generated detailed classification reports for each model, providing insights into precision, recall, and F1-score for each class. These metrics are importantly vital for medical diagnostic tests due to their implications for patient care. Our rigorous evaluation process ensured that we comprehensively assessed each classifier's predictive capabilities, highlighting our approach's strengths in accurately diagnosing glaucoma. The models and their accuracies and F1-Score are compared in Table 2.

The Random Forest algorithm (Alghamdi and Alharby, 2019), configured with 100 trees, gave us the best performance in the clinical data, whose architecture we can see in Fig. 7. Each tree in the Random Forest votes for either 'Glaucoma' or 'Healthy' based on its decision rules. The final classification is determined by majority voting, where the class that receives the most votes from the trees is chosen as the final prediction. This ensemble method enhances predictive accuracy and robustness by aggregating the results from multiple decision trees.

5.6. Final fusion model for glaucoma detection (multi-modal architecture)

The methodology we developed is illustrated in Fig. 8.

To develop a robust model capable of diagnosing glaucoma using both retinal images and clinical biomarkers, we implemented a fusion technique combining Vision Transformers (ViT, OWL-ViT, and Swin Transformer) and Convolutional Neural Networks (CNN) for image data with RandomForest for tabular clinical data. Our best hybrid model, the Swin Transformer hybrid model, leverages imaging and non-imaging data to enhance diagnostic accuracy to reach 99.4% on our testing dataset.

We used the Random Forest classifier to extract features from the tabular data. In this step, we trained the Random Forest on clinical biomarkers to identify patterns that can predict glaucoma. To ensure uniform data distribution, we standardized the processed tabular data using a StandardScaler. Then, we passed the extracted features through dense layers to create a dense representation.

The retinal images were processed using a dedicated preprocessing pipeline, which involved steps such as contrast adjustment and noise reduction to improve image quality. These preprocessed images were then fed into both the Vision Transformer and ResNet50 models to

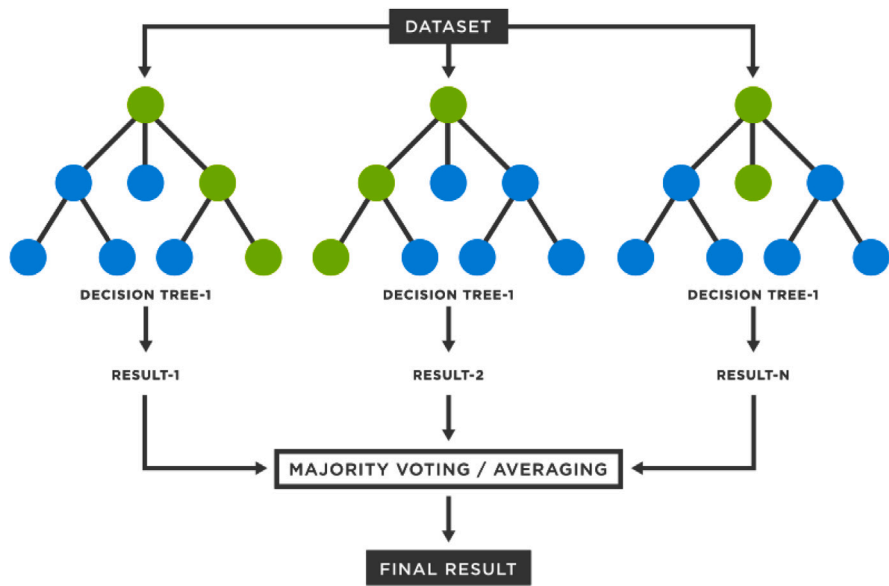


Fig. 7. Random Forest Architecture for Glaucoma Detection (Spotfire, 2023).

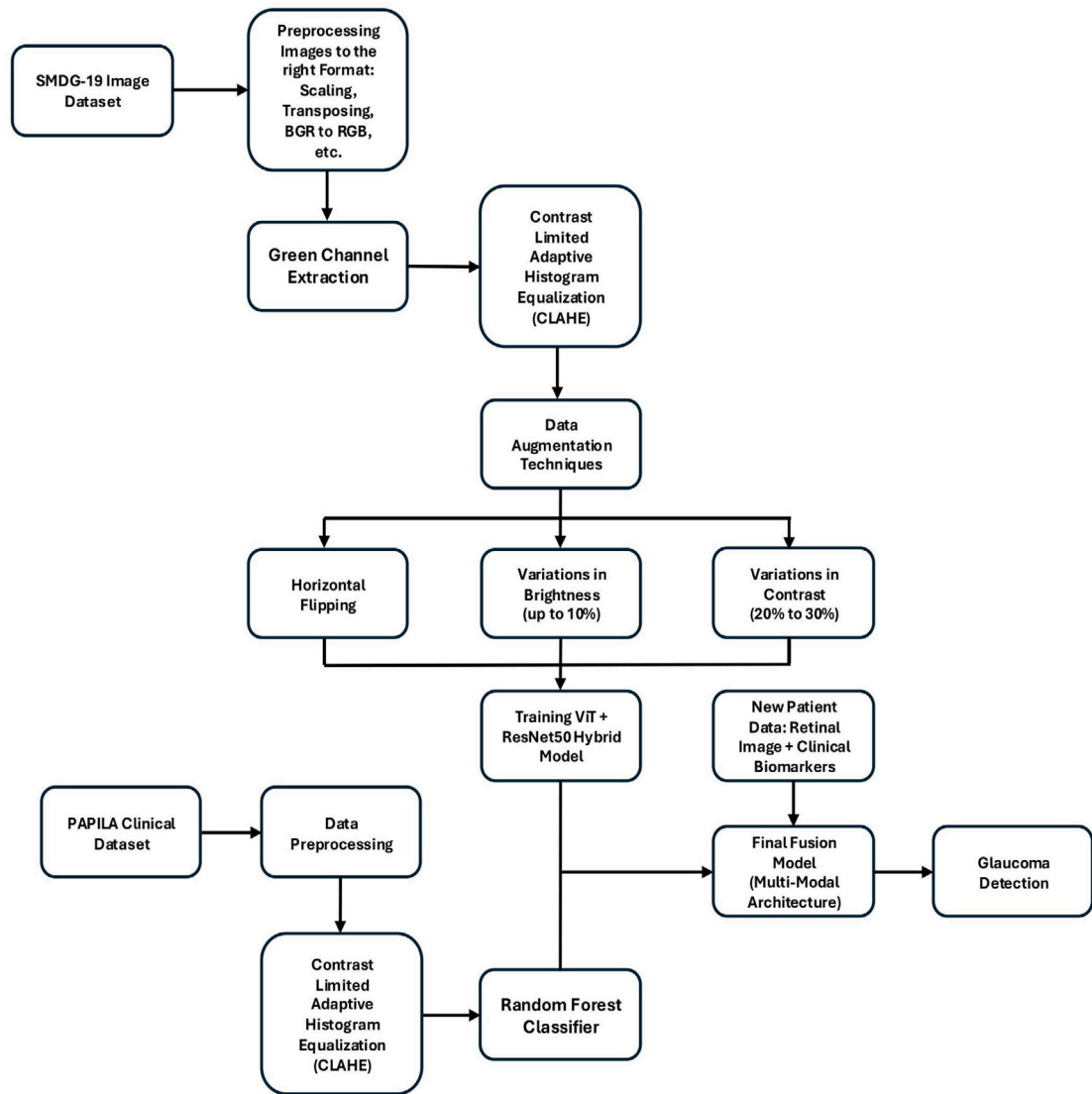


Fig. 8. Our Comprehensive Methodology.

extract feature representations of the retinal fundus images. The Vision Transformer divides the images into smaller patches and converts each patch into a feature vector using a self-attention mechanism, capturing global contextual information and intricate patterns across the entire image. At the same time, our ResNet50 model, known for identifying spatial hierarchies within images, extracts localized features that can recognize subtle and localized signs of glaucoma.

In our innovative hybrid model, we combined image features from the Vision Transformer and ResNet50 with clinical biomarkers extracted using the Random Forest Model. This fusion created a powerful feature vector integrating both image and clinical information at an early stage, enhancing the model's performance.

For classification, the combined feature vector was meticulously processed through a final dense layer with a softmax activation function, producing probabilities for the 'Glaucoma' and 'Non-Glaucoma' classes. Our model then makes classifications based on a threshold probability of 0.5.

The hybrid fusion model was meticulously compiled using the adaptive Adam optimizer, dynamically adjusting learning rates during training to optimize model performance. Furthermore, the loss function employed was the effective sparse categorical cross-entropy, perfectly suited for our classification task. Our model was rigorously trained on a comprehensive combined dataset, encompassing both retinal images and vital clinical data.

Our model does not just analyze images and biomarkers separately — it processes them simultaneously for high predictiveness and precise interpretation. By evaluating output probabilities, our model accurately diagnosis the presence or absence of glaucoma, significantly improving detection accuracy. This innovative approach also enables the model to recognize various pathological features in retinal images, potentially leading to more reliable and earlier detection of the disease.

5.7. Implementation details

We developed and implemented our hybrid model using a combination of high-level programming tools and advanced computing hardware. We utilized TensorFlow 2.x and Keras libraries to construct and train our deep learning models, taking advantage of their extensive support for neural network operations and GPU acceleration. For data preprocessing and analysis, we relied on the Pandas and NumPy libraries, which offered robust data manipulation capabilities necessary for handling and preparing large datasets. The visualization of the model's performance metrics, such as learning curves and confusion matrices, was accomplished using Matplotlib and Seaborn.

Our model was trained on the powerful Tesla 100 V GPU provided by Google Colab Pro, enabling efficient processing of large-scale medical image datasets. Our ViT model underwent 30 epochs of training for 5-6 h, resulting in a 87% test accuracy. In contrast, our OWL-ViT and Swin Transformer hybrid models, due to their complexity, achieved excellent test accuracies of 99.3% and 99.4% after just ten epochs of training. The OWL-ViT model took around 30 h to train, while the Swin Transformer only required 2 h, showcasing the efficiency of our computational setup in handling complex neural network calculations and large data volumes. This setup not only ensured the reproducibility of our results but also the practical feasibility of our approach for other researchers or practitioners in similar settings.

6. Experimental results and discussions

6.1. Results

We rigorously assessed the three fusion models we developed using a range of metrics to gauge their effectiveness in identifying glaucoma from retinal images and clinical data. These metrics encompass precision, recall, F1-score for binary classification tasks, accuracy for overall performance, and the Matthews correlation coefficient (Anbalagan et al., 2023), which we have found to be a superior

Table 3

Detailed classification report of the model performances for all three hybrid models.

Model	Class	Precision	Recall	F1-score
ViT + ResNet50	Non-glaucoma	0.94	0.85	0.89
	Glaucoma	0.80	0.92	0.85
	Final Test Accuracy	87.68%		
	MCC	0.754		
Fine-tuned OWL-ViT + ResNet50	Non-glaucoma	0.993	0.997	0.995
	Glaucoma	0.995	0.988	0.991
	Final Test Accuracy	99.3%		
	MCC	0.987		
Fine-tuned Swin Transformer + ResNet50	Non-glaucoma	0.995	0.995	0.995
	Glaucoma	0.992	0.993	0.993
	Final Test Accuracy	99.4%		
	MCC	0.988		

metric for evaluating our model's performance

Table 3 presents a comparison of the performance of all three hybrid models on the test dataset. It is evident that both the Fine-tuned OWL-ViT + ResNet50 model and the Fine-tuned Swin Transformer + ResNet50 model outperform the base ViT + ResNet50 model.

Our top-performing model is the Fine-tuned Swin Transformer + ResNet50 model. It achieved high precision and recall scores of 0.995 and 0.995, respectively, for the non-glaucoma class, indicating its reliability in making predictions for this category. Equally excellent precision and recall scores of 0.992 and 0.993, respectively, were achieved for the glaucoma class, demonstrating the model's ability to identify almost all positive cases. The F1-scores of 0.995 for non-glaucoma and 0.993 for glaucoma further validate the model's robustness by balancing precision and recall. The overall test accuracy of the model stood at 99.4%, confirming its efficacy in distinguishing between glaucoma and non-glaucoma cases. Furthermore, the Matthews correlation coefficient (MCC) (Anbalagan et al., 2023) of 0.988 indicates a strong correlation between the observed and predicted classifications.

The confusion matrix for our best model is presented in Fig. 9. This matrix effectively illustrates the predictive accuracy of our model. For the non-glaucoma class, the model identified 1746 true negatives, demonstrating its robust capability to accurately recognize non-glaucomatous conditions. Moreover, it only had 8 cases classified as false positives.

In the case of the glaucoma class, our best model impressively predicted 1111 true positives, affirming its high sensitivity and effectiveness in detecting glaucoma. The model had only 9 false negatives, which, although minimal, suggests the need for further refinement to capture more subtle manifestations of the disease.

Overall, these results indicate a high level of effectiveness in the model's ability to classify cases correctly, highlighting its usefulness in clinical settings for accurately detecting glaucoma. The areas identified for improvement, such as reducing false positives, provide actionable insights that can be addressed in subsequent enhancements to the model.

The training process was monitored for overfitting, with both training and validation losses and accuracies plotted over epochs. The plots of our best methodology have been shown in Fig. 10.

The training and validation accuracy curves closely track each other during the entire training process. This indicates that the model not only effectively learned from the training data but also retained its ability to generalize to new data. Additionally, the validation accuracy consistently mirrored the training accuracy, confirming that the model was not simply memorizing the training data but genuinely learning to

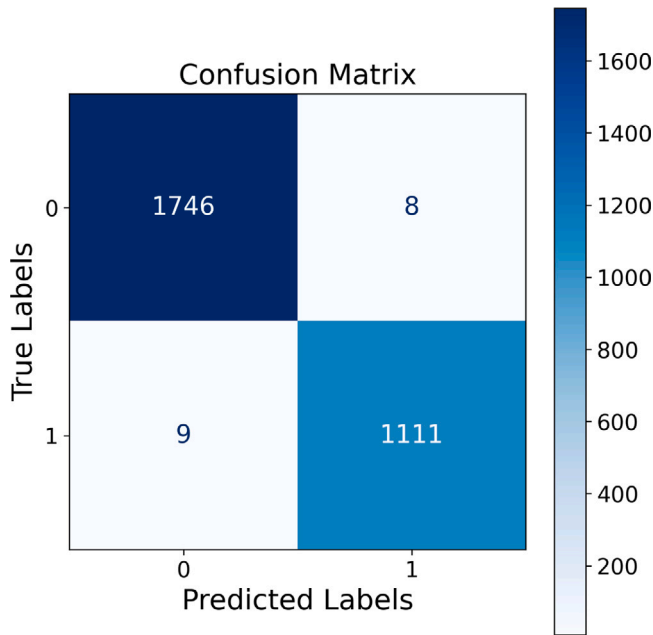


Fig. 9. Confusion matrix of our best model's predictions.

generalize from it. Similarly, the loss curves showed a steady decrease in both training and validation loss, which demonstrates the model's ability to improve its predictions over time without overfitting. The parallel decline of training and validation loss further supports the model's effective learning and generalization capabilities.

To ensure robust model performance and prevent overfitting, several strategies were implemented. A dropout rate of 0.5 was used to randomly deactivate a portion of the neurons during training, which helps prevent the model from becoming overly dependent on any single or small group of features. L2 regularization was applied to the weights of the neurons, penalizing large values and thus simplifying the model's complexity, which is a common technique to avoid overfitting.

The careful monitoring of these metrics and along with the strategic use of regularization and dropout techniques helped maintain a balance between learning the intricacies of the training data and generalizing well to new, unseen data. These results collectively demonstrate the high degree of precision and reliability of our fusion model, highlighting its potential as a powerful tool in clinical settings for the early detection of glaucoma.

6.2. Discussion

The effectiveness of our multi-modal glaucoma detection algorithm is rooted in its hybrid architectural design, which leverages the strengths of both the Vision Transformer and the ResNet50 convolutional neural network. This design choice directly addresses our first research question: whether the integration of Vision Transformers with traditional machine learning analysis of clinical biomarkers can enhance the accuracy and reliability of glaucoma detection compared to single-modality image analysis methods as we were able to achieve higher accuracy than existing image-based glaucoma detection methods. The model begins with a Vision Transformer, adapted to our specific image size and configured without the top layer to enable custom feature extraction. This transformer model is pre-trained, leveraging large dataset features to enhance its ability to discern intricate details in retinal images crucial for detecting glaucomatous changes.

Parallel to the ViT and OWL-ViT and Swin Transformer, we employed the ResNet50 model, renowned for its deep residual learning capabilities. Adapted to our image size and equipped with pre-trained

weights from ImageNet, ResNet50 ensures a broad spectrum of image features is captured. To maintain the integrity of the learned features and optimize computational efficiency, the layers of the ResNet50 model are set to non-trainable.

Both models feed into a shared input layer and process the same input images simultaneously. The features extracted by both the Vision Transformer and ResNet50 are flattened and concatenated, creating a comprehensive feature set that integrates global and local perspectives of the image data. This combined feature set then passes through several fully connected layers with GELU activation, chosen for its success in deep learning applications for providing non-linear transformation capabilities.

A dropout layer with a rate of 0.5 is strategically placed to prevent overfitting. This was accomplished by randomly omitting a subset of features during training, which helps the model generalize better. The architecture concludes with a softmax activation layer that outputs the probabilities for two classes, representing the presence or absence of glaucoma.

The model is compiled with the AdamW optimizer, which extends the traditional Adam optimizer by incorporating weight decay for enhanced regularization. This decision is especially effective in controlling the first and second moments of gradients, facilitating more stable and consistent training outcomes. The learning rate is set at 0.0001 to facilitate gradual and steady learning. The model employs sparse categorical cross-entropy as the loss function, suitable for our classification task.

After extensive training, the ViT+ResNet50 model underwent 30 epochs, while both the OWL-ViT+ResNet50 and Swin Transformer+ResNet50 models underwent 10 epochs due to their complex structures. The best performing model was the Swin Transformer + ResNet50, which achieved a final training accuracy of 99.8% and test accuracy of 99.4%. The fusion model attained an F1-score of 0.993 for glaucoma and 0.995 for non-glaucoma, correctly classifying 2,857 out of 2,874 test images. These results demonstrate the model's robust learning and its ability to generalize across different datasets.

Our approach outperforms existing models, as demonstrated in Table 4, with significant improvements in F1-Score and Test Accuracy. The Table outlines a comprehensive comparison of our hybrid models against previous studies, utilizing a variety of datasets and classification approaches. Our fine-tuned Swin Transformer + ResNet50 model achieved an exceptional accuracy and F1-score of 99.4%, surpassing other methods reported in recent studies. For instance, the InceptionV3 + ResNet50-based U-Net segmentation previously led with an accuracy of 98.97% and an F1-score of 99.42%. Similarly, our fine-tuned OWL-ViT + ResNet50 model also showed outstanding performance, with both accuracy and F1-score at 99.3%, highlighting the effectiveness of integrating advanced Vision Transformer architectures with conventional CNNs like ResNet50. This methodological approach effectively addresses our second research question by demonstrating how the proposed multi-modal diagnostic tool performs against traditional and current AI-based methods in terms of accessibility, accuracy, and cost-effectiveness.

Unlike many existing methodologies that primarily rely on imaging data for glaucoma detection, our approach integrates a multi-modal dataset containing both retinal images and clinical biomarkers. Our models were also trained and tested on a dataset that combines images from 15 different datasets, thus showing the accuracy of our model in providing a solution that is generalized across different kinds of images. Our comprehensive approach takes into account not only the visual indicators present in the images but also crucial clinical data such as patient age, intraocular pressure, and other relevant health biomarkers. By combining these diverse data sources, our model offers a holistic view of the patient's condition, enabling more accurate and confident diagnosis. This integration of clinical biomarkers with image data is particularly important in cases where visual symptoms might be subtle or not entirely indicative of the disease, thus enhancing the

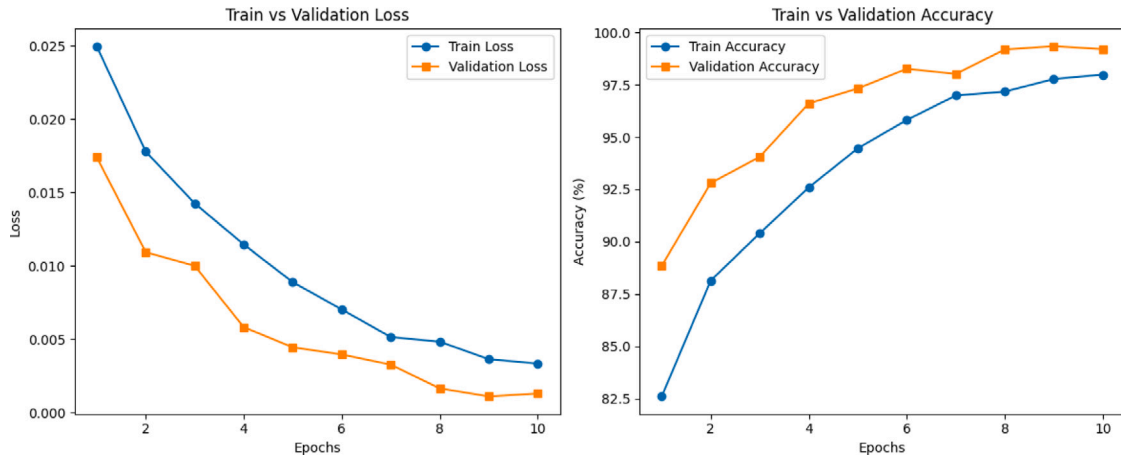


Fig. 10. Confusion matrix of our best model's predictions.

Table 4

Comparison of various studies on glaucoma detection using different datasets and methods.

Study	Dataset	Classification	Accuracy %	F1-score %
Kashyap et al. (2022)	LAG	Densenet-201 + U-Net segmentation	96.90	96.28
Kim et al. (2019)	Private Dataset	ResNet-152	93.5	92.9
Elmoufidi et al. (2022)	RIM-ONE, REFUGE	CNN + BEMD	99.17	–
Shoukat et al. (2023)	RIM-ONE, ORIGA, GI020, DRISHTI-GS1	ResNet-50	96.15	97.85
Shyamalee et al. (2024)	RIM-ONE, ACRIMA, ORIGA, REFUGE, DRISHTI	InceptionV3 + ResNet50-based U-Net segmentation	98.97	99.42
Our Work	SMDG-19 (all datasets combined)	ViT + ResNet50	87	85.32
Our Work	SMDG-19 (all datasets combined)	Fine-tuned OWL-ViT + ResNet50	99.3	99.3
Our Work	SMDG-19 (all datasets combined)	Fine-tuned Swin Transformer + ResNet50	99.4	99.4

predictive power and reducing dependency on single-modality data, which is often susceptible to higher rates of diagnostic errors. Hence, the results achieved through our methodological innovations provide clear evidence that the research questions guiding this study have been effectively addressed. The hybrid model's robust performance across diverse datasets not only demonstrates its practical utility in clinical settings but also represents a significant advancement in the field of glaucoma detection.

7. Conclusions and future directions

7.1. Conclusions

Our study contributes to the field of glaucoma detection by creating a hybrid fusion model that integrates retinal imaging with clinical biomarkers, employing a novel combination of advanced machine learning techniques. The motivation behind our research was the need for improved and early detection methods for glaucoma, a leading cause of blindness worldwide. Traditional diagnostic methods that rely solely on imaging data often lack the ability to provide a comprehensive and accurate prediction, especially when considering other health conditions of the patient.

Our approach involves the use of advanced machine learning techniques, specifically a blend of Vision Transformers and Convolutional Neural Networks with clinical data. This establishes a robust framework that enhances diagnostic accuracy and has the potential to result in better patient outcomes through earlier and more precise interventions.

To the best of our knowledge, this is the first work that combines multiple Vision Transformers such as ViT, OWL-ViT, and Swin Transformer with ResNet50 architecture. Our methodology also performs clinical biomarker analysis for more accurate glaucoma detection. This unique approach captures both macro and micro-level details within retinal images and incorporates valuable clinical data, resulting in a more comprehensive and accurate diagnostic model. The Transformer processes images as sequences of patches, applying self-attention mechanisms to highlight intricate patterns, while the ResNet50 model excels in identifying spatial hierarchies in the images, recognizing subtle and localized signs of glaucoma.

Our methodology involves extracting tabular features from clinical data using the RandomForest algorithm. We then performed feature-level early fusion with image features extracted from the hybrid Vision Transformer + ResNet50 model. This comprehensive technique, combining both imaging and non-imaging data, allows for a more accurate diagnosis of glaucoma.

The models presented here were trained on a combined dataset of both retinal images and clinical biomarkers. We achieved significant improvements in diagnostic accuracy, demonstrating the potential of our approach in clinical practice. Unlike existing methodologies, our approach uniquely combines cutting-edge deep learning models with clinical data analysis, overcoming the limitations of relying solely on imaging data. This hybrid model architecture has not been previously applied to glaucoma detection, setting our work apart and offering valuable insights into the realm of automated medical diagnosis.

The advancements mentioned not only improve clinical outcomes but also provide a scalable solution that could be deployed in diverse

healthcare settings to alleviate the global burden of glaucoma. Our findings indicate significant advancements over traditional single-model approaches, creating a robust framework that enhances diagnostic accuracy and facilitates earlier, more precise interventions.

The hybrid model we are using has many advantages, but it also has some limitations that must be acknowledged. Integrating Vision Transformers with Convolutional Neural Networks demands substantial processing power due to its complexity. Moreover, the model performance heavily relies on the quality and diversity of the input data. Inconsistencies in data quality, particularly in terms of image resolution and the comprehensiveness of clinical biomarker information, could lead to variability in diagnostic accuracy. This sensitivity to data quality underscores the need for rigorous data preprocessing and continuous updates and calibrations of the model as new data becomes available. Furthermore, the reliance on high-quality data emphasizes the importance of robust data collection protocols, which can pose challenges in fragmented or non-standardized medical data collection environments.

We further plan to tackle these limitations by diving into longitudinal data analysis. We are actively working to validate our methodology by collaborating with private clinics, and we are in communication with them. Additionally, we have developed a user-friendly website where we will soon launch our model. This will create a dynamic platform for users and healthcare providers to access our innovative methodology.

7.2. Future directions

The hybrid model we have developed has shown great effectiveness in automating the detection of glaucoma. By fusing Vision Transformers with ResNet50, our system not only streamlines the detection process but also ensures high accuracy and sensitivity in identifying glaucomatous changes. This capability is vital for enhancing screening processes and alleviating the workload of healthcare professionals, especially in regions with limited access to ophthalmological expertise.

We are taking our work to the next level by launching our model on a user-friendly website. This will create an accessible platform for healthcare providers to utilize our improved diagnostic tool in real clinical settings. Our online platform will offer real-time analysis of retinal images, providing immediate results that can empower and expedite treatment decisions. Our team is in communication with medical experts to validate our model based on practical feedback, ultimately enhancing its reliability and accuracy in diagnosing glaucoma. Furthermore, we are dedicated to boosting the accuracy of our model by integrating longitudinal data analysis.

CRediT authorship contribution statement

Rishikesh Sivakumar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anita Penkova:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data will be shared upon reasonable request.

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